DESIGN SYSTEMS

TECHNICAL DATA SHEET LIVIUS-A
HIGH-PERFORMANCE NON-SHRINK SELF-LEVELLING MORTAR

DESCRIPTION

Ecological self-levelling mortar based on geo-binder, coloured and ready to use, for the construction of exposed finished floors subject to medium/heavy or pedestrian traffic, wear-resistant, without Portland cement.

TECHNICAL CHARACTERISTICS

Self-levelling

LIVIUS-A is a self-levelling mortar based on geo-binder. The product has high mechanical strength and the emissions linked to its production are 50% lower than those of a similar product made with Portland cement.

Timeless design

With its textured, essential appearance, similar to that of concrete, it is the ideal material for creating floors in commercial and industrial settings, giving interiors a timeless design. Its technical characteristics, combined with its textured, essential appearance, make it possible to create durable floors with LIVIUS-A that are resistant both mechanically and to abrasion.

Non-Newtonian fluid

Thanks to its exclusive nanoclay compound, LIVIUS-A is a non-Newtonian fluid: its viscosity varies according to the shear stress applied, thus allowing it to adapt to any desired shape.

- Decorative and industrial self-levelling mortar.
- Non-Newtonian fluid.
- High abrasion resistance.
- High mechanical strength.
- For indoor use.
- Can be ground.
- Can also be used on substrates with underfloor heating coils.
- Can be applied at temperatures from +5°C to +35°C.
- Available in coloured version.
- Applied in thicknesses from 4 to 40 mm.

FIELDS OF USE

- Thanks to its high mechanical and abrasion resistance, LIVIUS-A can be left exposed as a finished floor; it is ideal for residential, commercial or industrial use and for areas with heavy traffic, and is resistant to thermal shock and frost.
- Ideal in residential settings.

- Good aesthetic quality and colour rendering.
- Mechanical, chemical, food and textile plants.

WORK PHASES

- The substrates must be dry, solid, free from dust, friable or loose parts, paints, waxes, oils and any other type of contaminant.
- The substrate must have a minimum compressive strength of 25 N/mm^2 and a tensile strength of at least 1,5 N/mm^2; in any case, the mechanical strength of the substrate must be suitable for the type of use and the loads expected on the floor.
- The moisture content of the substrate must not exceed 7%. On surfaces with moisture above 7%, the three-component epoxy-cement primer Tix-Cem must be applied beforehand; in any case there must be no capillary rising damp.
- Before starting the preparation of LIVIUS-A, a special strip of compressible material must be applied along the perimeter of the rooms to be floored and around any vertical elements present (pillars, columns, etc.).
- Concrete, ceramic or natural stone surfaces must be prepared mechanically by grinding with diamond discs, shot-blasting or sanding, depending on the characteristics of the substrate. A bonding primer must then be applied: UNI-PRIMER 280 NEW for cementitious substrates, Primer A300 for low-absorption substrates (ceramic, porcelain stoneware, stone, etc.), three-component epoxy-cement Primer Tix-Cem for surfaces with moisture above 7%; all of this reinforced if necessary with Rete 320 (alkali-resistant glass-fibre mesh). On the still-wet primer, broadcast Quartz 1,2 to refusal. Before pouring LIVIUS-A, remove the excess sand by vacuuming.
- On the prepared substrate, mix LIVIUS-A with 14-18% water and spread it in a single layer, with a thickness of between 4 and 40 mm, using a spreader bar, never going back over material already applied.
- The minimum application thickness is 4 mm for decorative floors and 1 cm for high-performance floors (industrial type).
- It is important that the pour is carried out continuously, without interruptions, to prevent the floor from showing evident levelling defects and marked colour inconsistencies.
- During application, respect all expansion joints existing in the substrate.
- The quantity of LIVIUS-A to be prepared each time must be that which can be used within 15 minutes at a temperature of +23°C; the working time varies with temperature and decreases as it rises.
- Do not use LIVIUS-A on wet, metal, rubber, wood or PVC surfaces, or on non-cementitious waterproofing membranes.
- Do not use LIVIUS-A at temperatures below +5°C or above +35°C.
- Approximately 3 days after the application of LIVIUS-A, the surface must be protected with a finish from the Design Systems range: DS-CS, DS-SILC, EPOX, DS-POLTEC.

TYPICAL PRODUCT DATA

DESCRIPTION	RESULT
Regulatory compliance	N.C.

Colour / consistency	Can be coloured / powder
Solids content	100%
Bulk density	Approx. 1 kg/l
Hazard classification (Directive 1999/45/EC)	Irritant: read carefully the instructions given in this data sheet, on the bag and in the safety data sheet
Storage	6 months in a dry, ventilated place
Packaging	25 kg
Customs tariff code	3824 50 90

FINAL PERFORMANCE VALUES

DESCRIPTION	RESULT
Compressive strength at 28 days (EN 13892-2)	≥ 80 MPa
Flexural strength at 28 days (EN 13892-2)	≥ 10 MPa
Resistance to solvents	Excellent
Resistance to oils	Excellent
Resistance to alkalis	Excellent
Resistance to acids	Fair
Reaction to fire	A2-s1,d0 – classified as non-combustible
Taber abrasion, weight loss (H22 wheel, 500 g, 200 cycles)	0,5 g
Freeze-thaw (300 cycles)	Retains over 80% of its performance

APPLICATION VALUES AND INFORMATION (AT 23°C AND 50% RH)

DESCRIPTION	RESULT
Mixing water	Approx. 14-18% (± 1%) of water per one part of LIVIUS-A
Working time of the mix	20 minutes
Consumption	1,8 kg/m^2 per mm of thickness
Application temperature	From +5°C to +35°C
Initial setting	Approx. 50 minutes
Foot traffic	Approx. 4 hours
Ready for service	7 days
pH of the mix	Approx. 12
Density of the mix	2,2 kg/l
Applicable thickness	From 4 to 40 mm

DESIGN SYSTEMS

The technical data given in this bulletin were obtained using standardised tests and standards, or modifications thereof, according to Design Systems systems. The methods applied may be requested from our technical service. The contents of this technical bulletin represent the best of our knowledge and experience. However, for uses other than those indicated, preliminary tests are recommended. In any case, Design Systems Srl accepts no responsibility for damage or defects caused by the use of its products, since the conditions of use are not under its control. It also advises that its technical service is available to customers for any information concerning the correct use of its products.

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